

EXPERIMENT 1:

OBJECTIVE: To study Lancashire Boiler

APPARATUS/MATERIALS USED: Model of Lancashire Boiler.

THEORY:

Working principle:

Lancashire boiler is very widely used as a stationary boiler because of its good steaming quality and because it can also burn coal of inferior quality. This boiler has two internal furnaces or flue tubes instead of one. The Lancashire boiler has cylindrical shell usually from 2 to 3 metres in diameter and from 8 to 9.5 m long. The main features of the Lancashire boiler and its brickwork setting are shown. The boiler consists of a cylindrical shell with flat ends, and two furnace tubes pass right through this. There are two side flues and one bottom flue formed by brickwork setting. These flues provide part of the heating surface on the main external shell. Here, the gases from the furnace pass to back end of the boiler where they dip (go down) and enter the bottom flue and travel through it to the front end of the boiler. The gases then divide themselves here and enter the side flues and travel through them to the back end of the boiler and then to the chimney.

This type of boiler is often fitted with cross-tubes called galloway tubes, as in the. They are fitted across the furnace tubes. They provide an increased heating surface and improve the circulation of the water. The boiler is provided with a blow-off cock at the bottom of the front end, and a feed check valve with a feed pipe on the front end plate.

Components:

The **mountings** usually provided on the boiler are : pressure gauge, water level gauge (or water level indicator), stop valve, safety valve, high-steam and low-water safety valve, fusible plug, and anti-priming pipe.

The **accessories** provided in large size boiler of this type are : superheater, economiser, and feed pump.

Some Lancashire boilers are fitted with shortened furnace and a number of smoke tubes. These smoke tubes increase the heating surface of the boiler. Lancashire boilers fitted with smoke tubes are known as multi-tubular boilers. They are also sometime called economical boilers.

Advantages and Disadvantages :

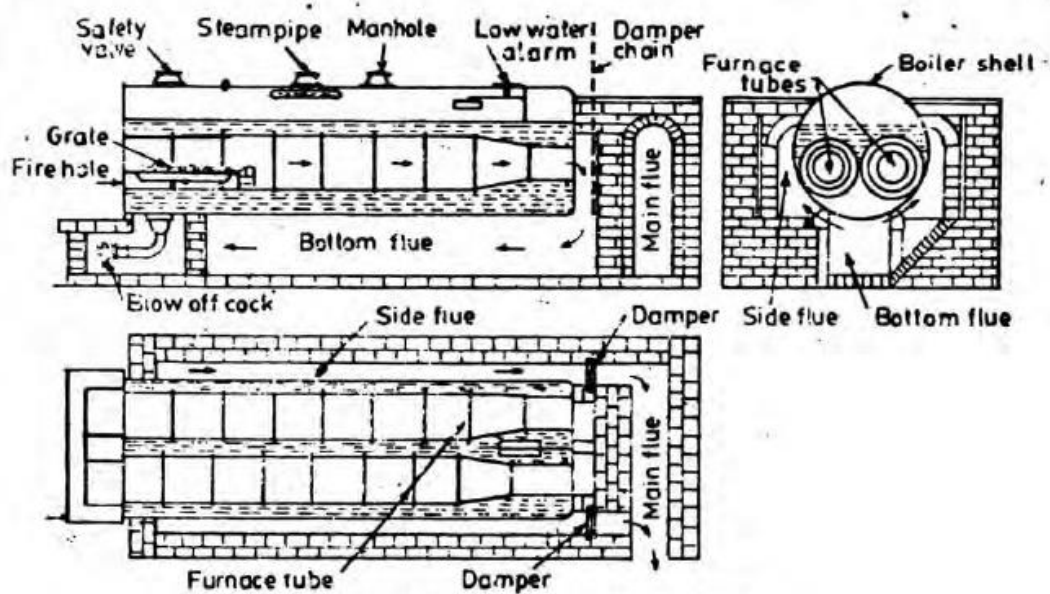
The features which have made the Lancashire boiler so popular are as under :

- Simple in design - there is little to go wrong.
- Ease of operation - it will put up with rough treatment.
- The ratio of the volume of the boiler to its rated evaporative capacity is high, consequently it is able to meet heavy peak loads without very great variation in steam pressure.

The disadvantages of Lancashire boiler are:

- The shell construction restricts the maximum working pressure to about 17.5 bar.
- It occupies considerable floor space.
- There is so large a water capacity and so little encouragement to water circulation, especially between the furnace tubes and the bottom of the shell, that it is impossible in emergency to raise steam pressure rapidly from the cold water.
- Brickwork setting is expensive in the first cost and troublesome in maintenance.
- The grate area is restricted by the diameter of the internal furnace tubes or flue tubes.

Construction of the boiler:



OBSERVATIONS:

[students need to attach front view and side view photos of the ACTUAL Boiler model and Label the parts properly]



Fig.: Front View of the boiler



Fig.: Side view of the boiler